



RELISH: LLM REgression with a Latent Iterative State Head

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[Paper & Code](#)



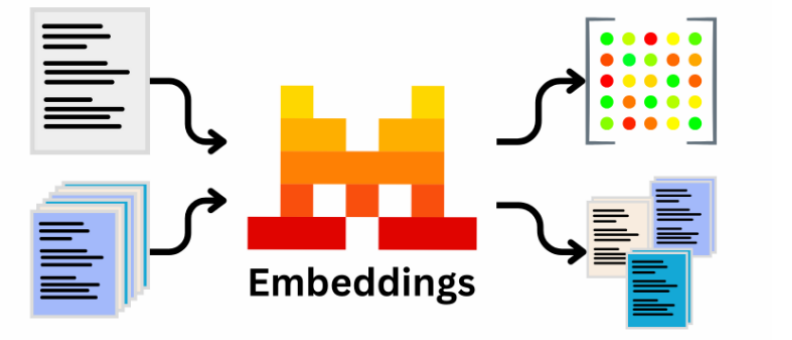
WHAT: RELISH is a lightweight text regression architecture that predicts scalar values directly from frozen LLM representations.
HOW: RELISH iteratively refines a learned latent state via cross-attention over token-level representations, then maps the final state to a point estimate with a linear regressor.
HOW WELL: Over 6 datasets, 4 LLMs, & 2 LLM training regimes, RELISH tops baselines from all 3 major LLM regression families while using only ~3.4–3.7M trainable parameters.

Motivation & Setup

How to adapt token-generating LLMs for continuous text regression tasks?

Semantic Similarity

How close two text artifacts are to each other in meaning?



- STS-Benchmark
- Sentences Involving Compositional Knowledge

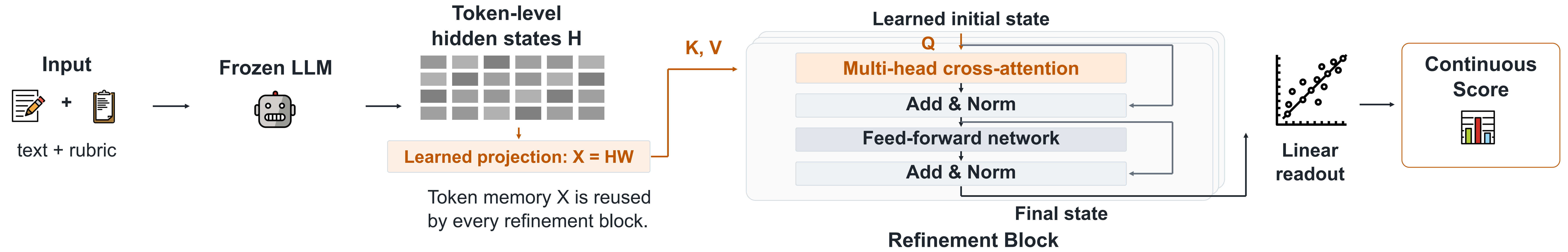
Quality Estimation

How good is a translation or a piece of code?



- Translation: WMT2020 English-to-Chinese, Russian-to-English, and Sinhala-to-English
- Code: StatLLM

RELISH iteratively distills token-level hidden states into a single latent state optimized for scalar prediction.

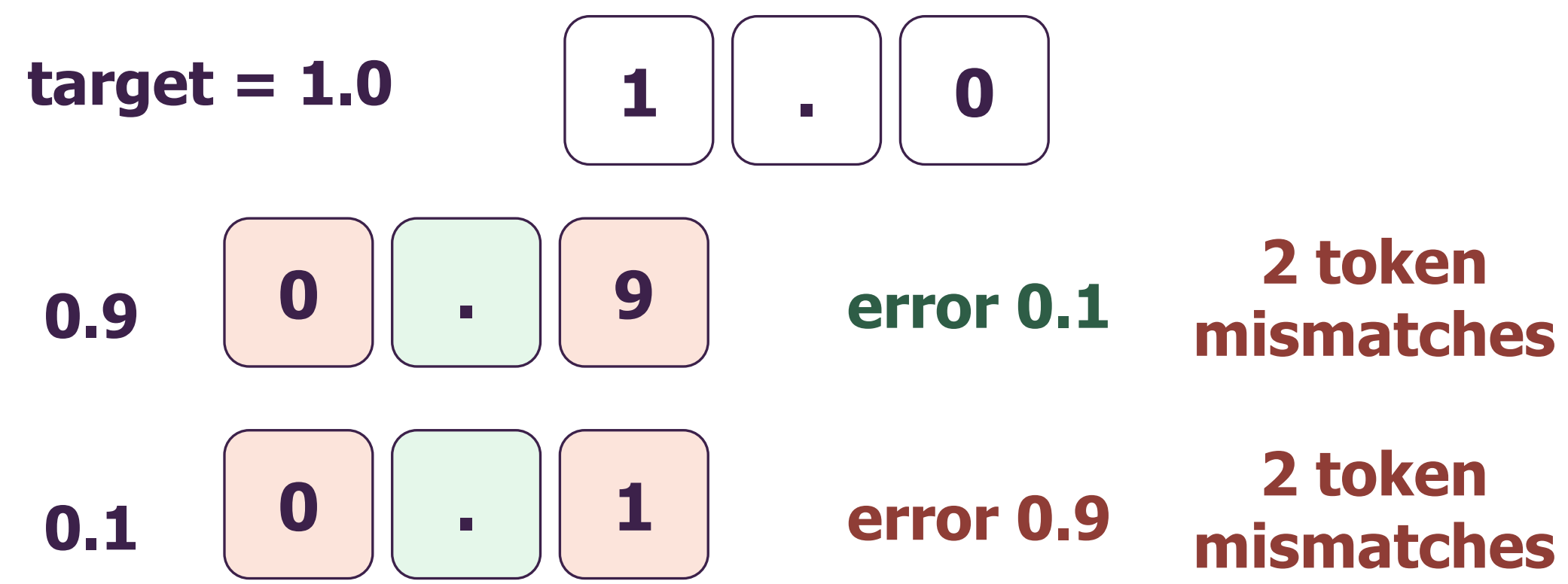


Limitations of Existing LLM-based Text Regression Methods

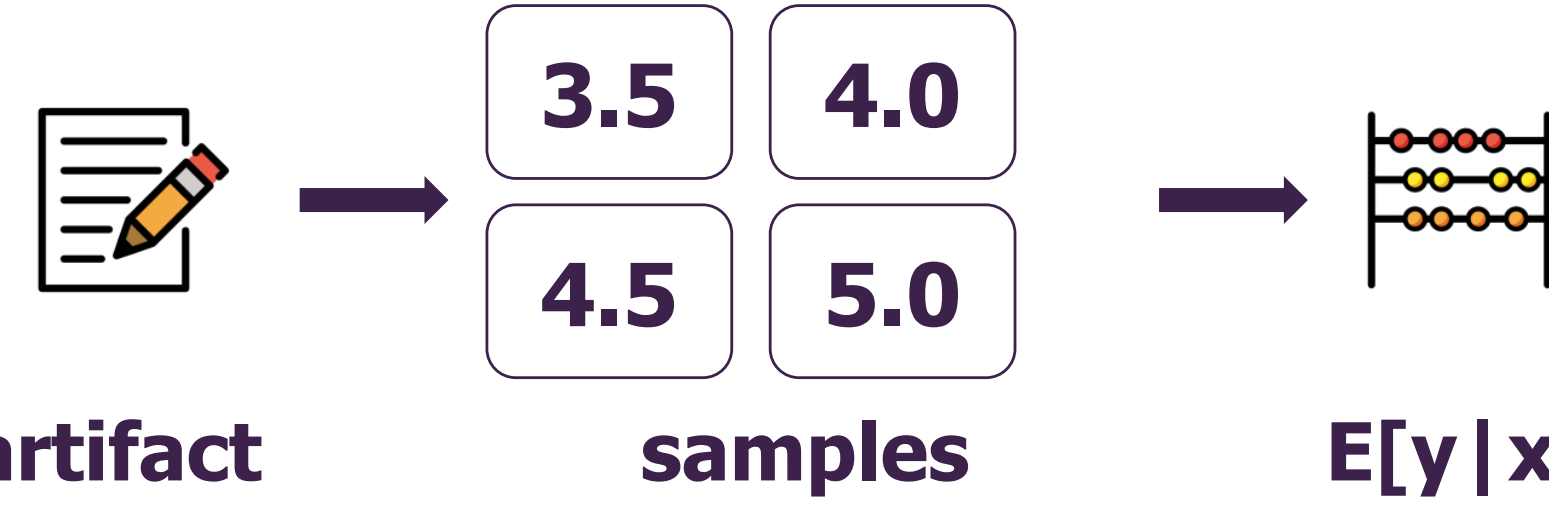
Language Modeling



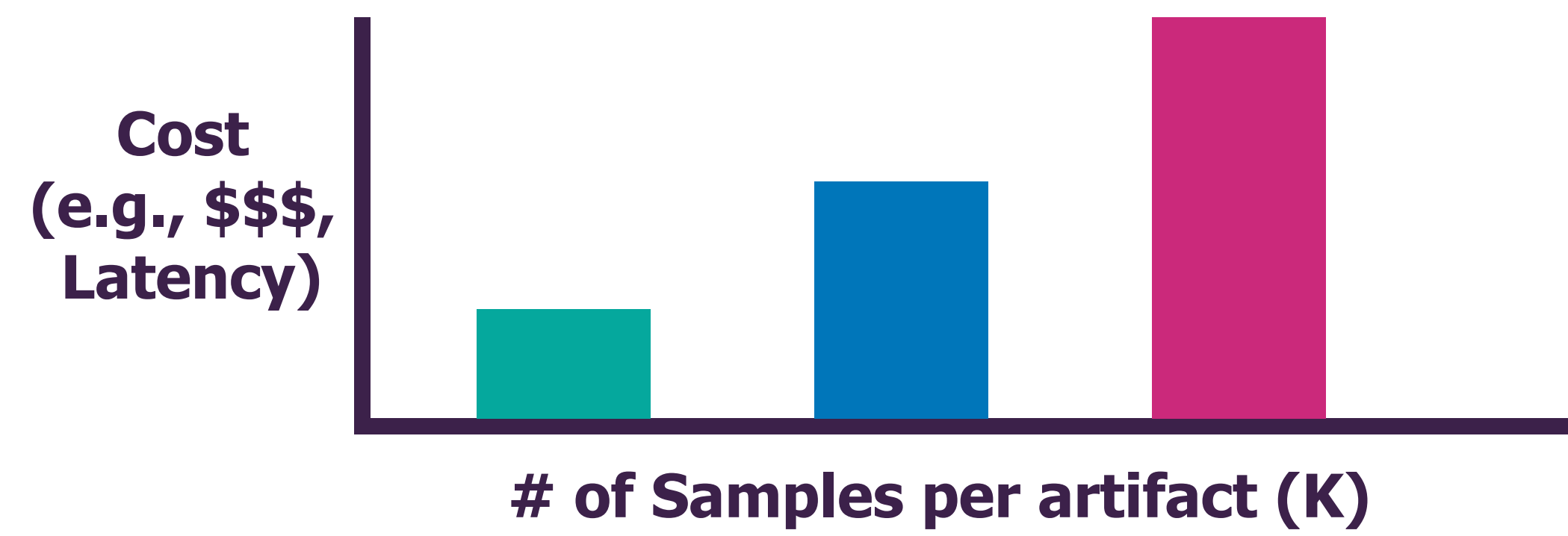
GAP: token likelihood does not directly penalize numerical distance.



Regression-aware Inference



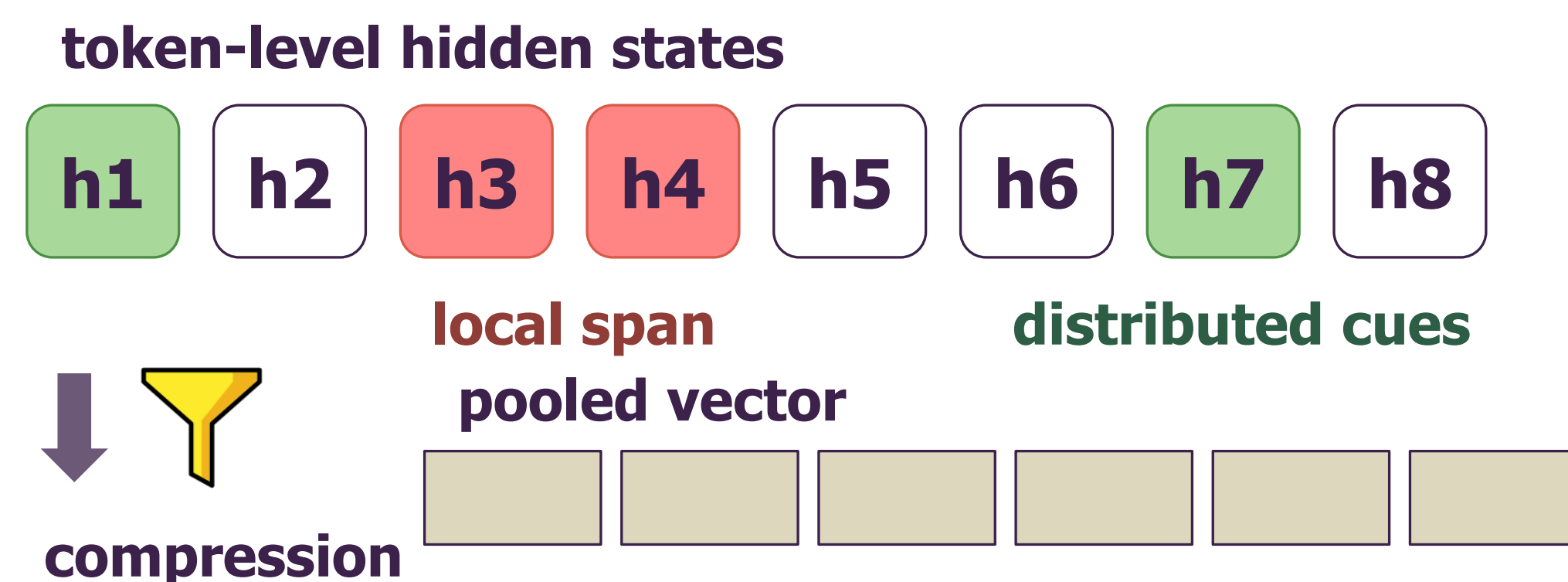
GAP: generating multiple samples per input increases inference cost.



Predictive Heads



GAP: fixed pooling can dilute local or distributed evidence.



Predictive Performance & Efficiency Results

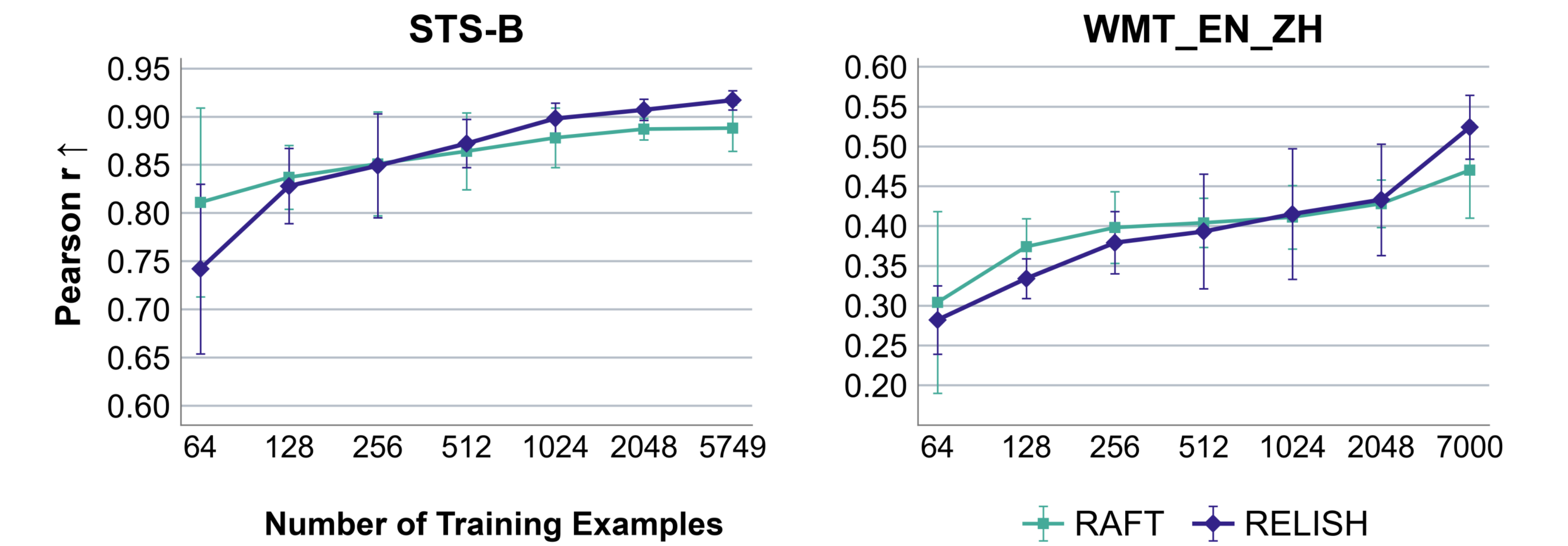
Metric	Unsupervised			Supervised							Ours
	Zero	Many	RAIL	Lin.	MLP	L+Zero	L+Lin.	L+MLP	L+RAIL	RAFT	RELISH
$r \uparrow$	55.0 $\pm$ 0.3	52.3 $\pm$ 0.8	56.0 $\pm$ 0.1	44.4 $\pm$ 0.0	56.3 $\pm$ 0.4	57.2 $\pm$ 0.4	44.9 $\pm$ 0.0	55.9 $\pm$ 0.5	61.8 $\pm$ 0.5	66.1 $\pm$ 0.3	72.9 $\pm$ 0.1
$\rho \uparrow$	55.5 $\pm$ 0.2	52.6 $\pm$ 1.0	58.8 $\pm$ 0.1	42.5 $\pm$ 0.0	53.7 $\pm$ 0.4	56.1 $\pm$ 0.5	43.0 $\pm$ 0.0	53.4 $\pm$ 0.3	62.9 $\pm$ 0.6	65.3 $\pm$ 0.5	71.4 $\pm$ 0.3
NRMSE $\downarrow$	25.1 $\pm$ 0.0	22.2 $\pm$ 0.5	25.5 $\pm$ 0.0	23.1 $\pm$ 0.0	17.8 $\pm$ 0.3	18.9 $\pm$ 0.1	23.5 $\pm$ 0.2	17.7 $\pm$ 0.1	17.9 $\pm$ 0.1	15.7 $\pm$ 0.2	12.6 $\pm$ 0.2

Pearson $r \uparrow$
+6.8 points
(+10.3% vs. RAFT)

Spearman $\rho \uparrow$
+6.1 points
(+9.3% vs. RAFT)

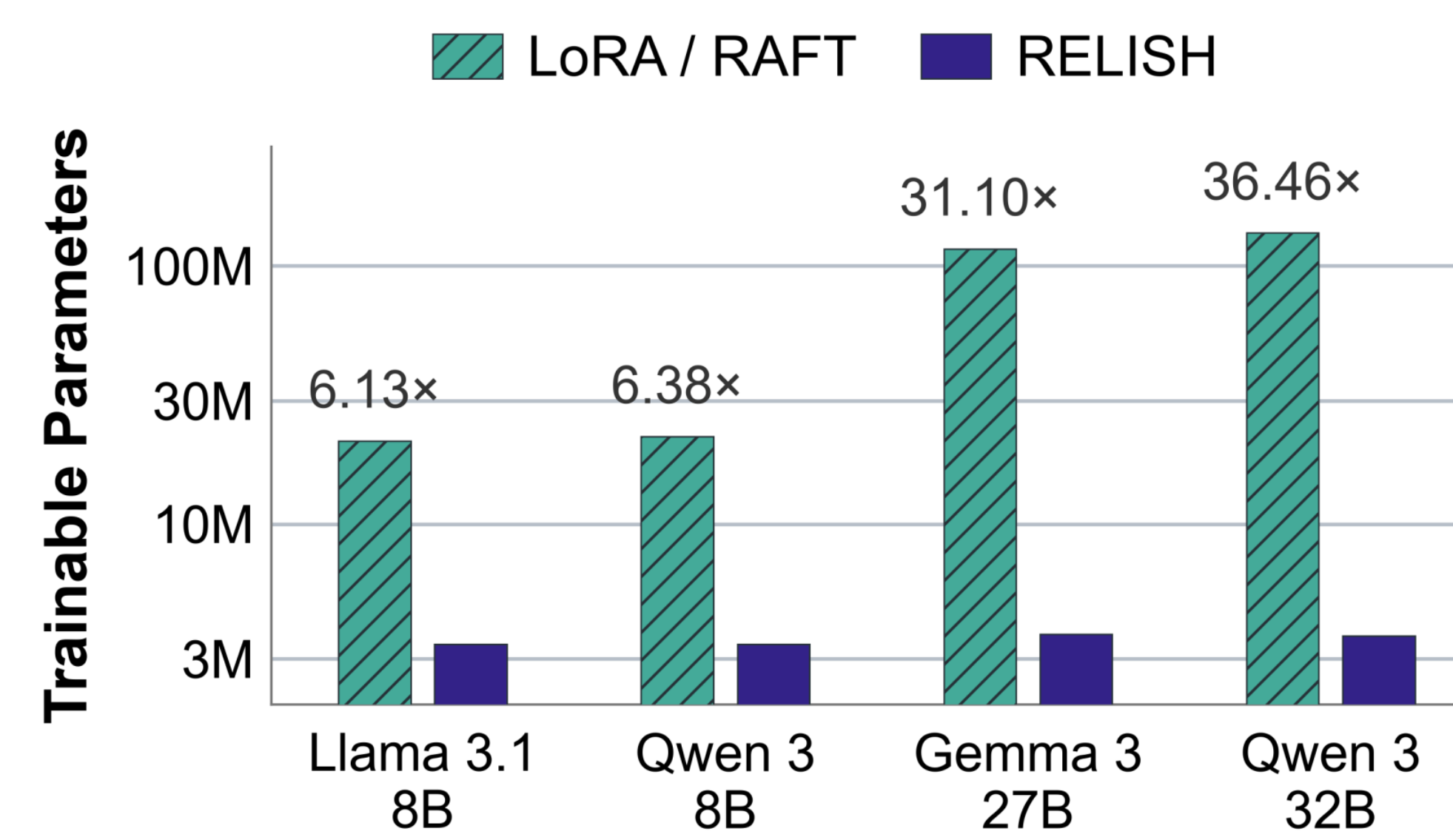
NRMSE $\downarrow$
-3.1 points
(-19.7% vs. RAFT)

Training data efficiency



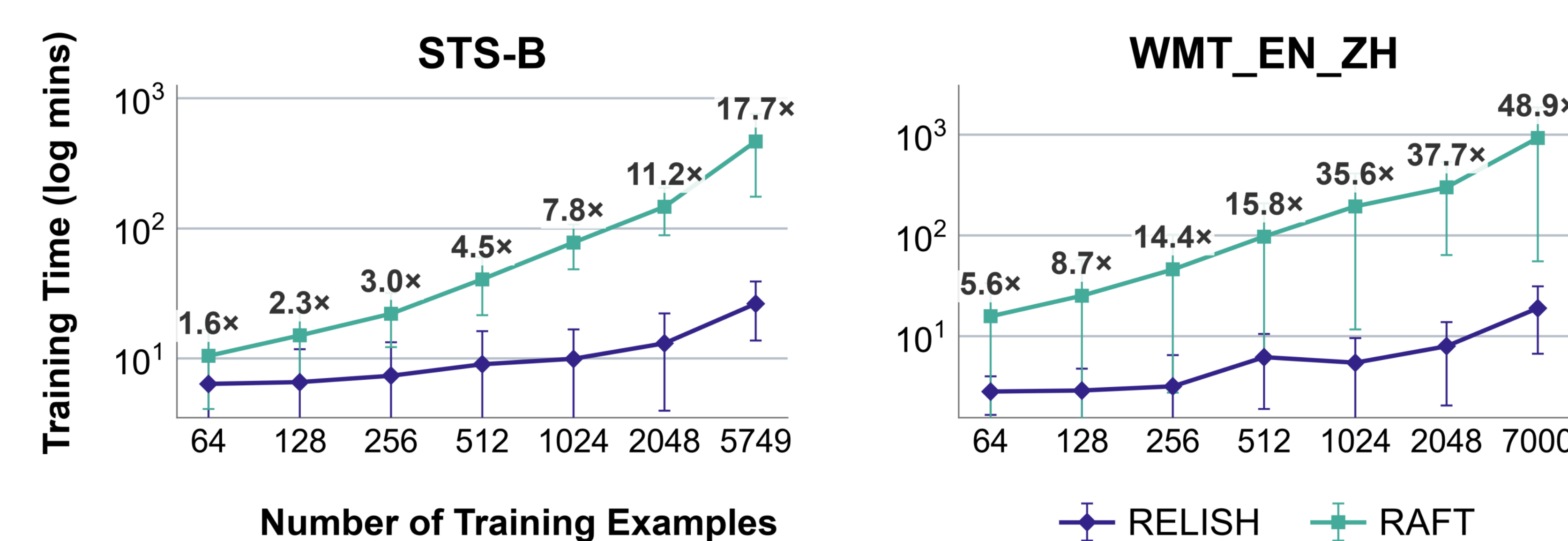
RAFT excels with ≤ 256 examples; **RELISH** surpasses **RAFT** on both datasets with $\geq 1,024$ training examples.

Trainable parameters



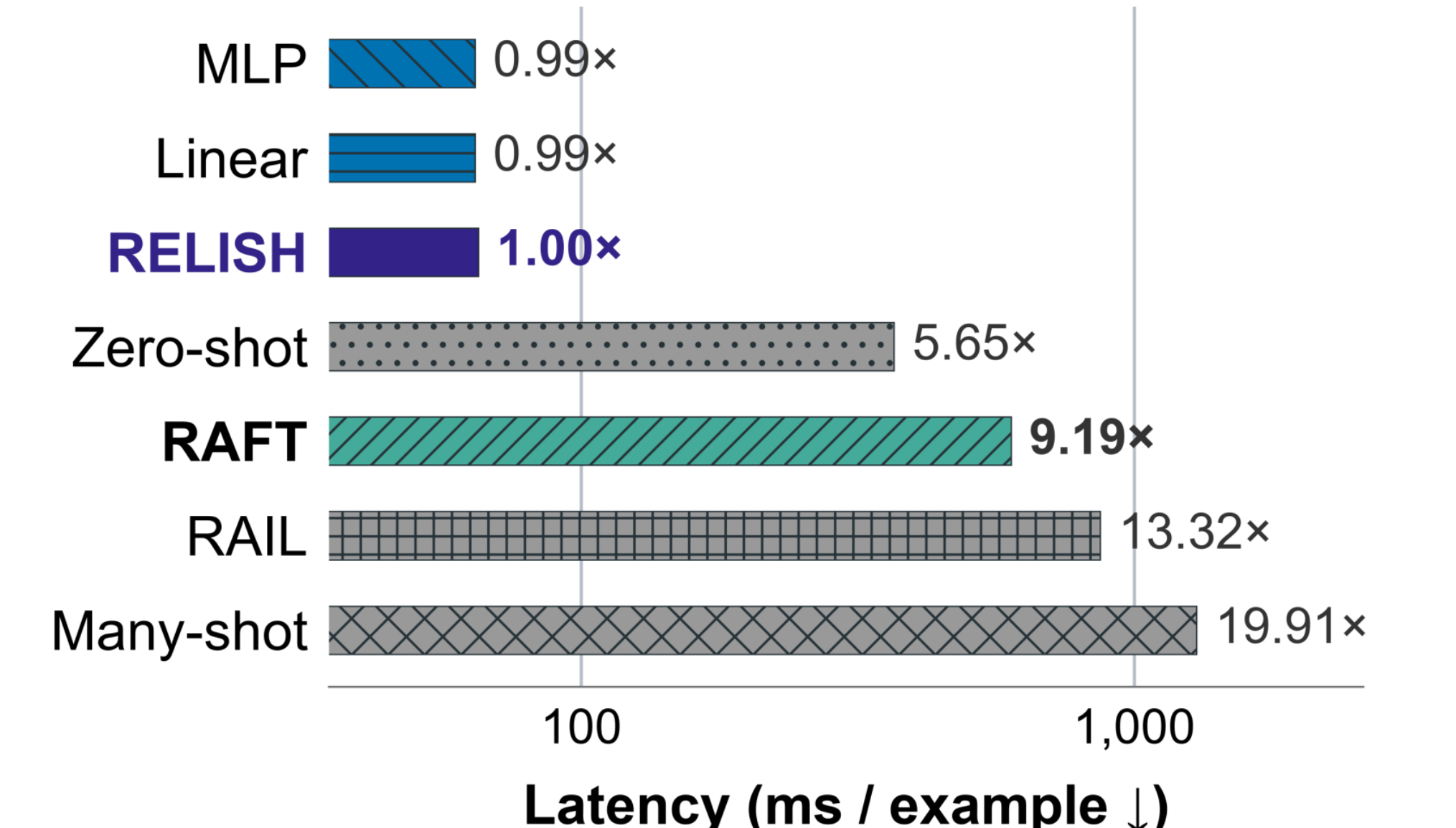
RELISH requires 6.13–36.46× fewer trainable parameters than RAFT.

Training overhead



RELISH fine-tunes 1.6–48.9× faster than RAFT across various training-data scales.

Inference latency



RELISH adds negligible (~1%) inference overhead vs. simple predictive heads.